

AI for Compliance

# BISfit

BIS Standards Recommendation Engine  
for Indian Micro & Small Enterprises



**BIS STANDARDS**  
MSME RAG System

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# 01 | Problem Statement

## The MSE Compliance Gap

- > 63 million MSEs in India face complex BIS compliance requirements
- > Weeks spent manually identifying applicable standards
- > 500+ BIS standards in Building Materials alone
- > Non-compliance risks: penalties, market rejection, safety failures
- > No affordable, accessible tool exists for smaller enterprises

**6–8 Weeks**

Average time to identify applicable BIS standards

**63M+**

MSEs in India needing compliance guidance

**₹2L–5L**

Estimated cost of non-compliance per incident

# BISfit — Ultra-Fast Standards Intelligence Solution

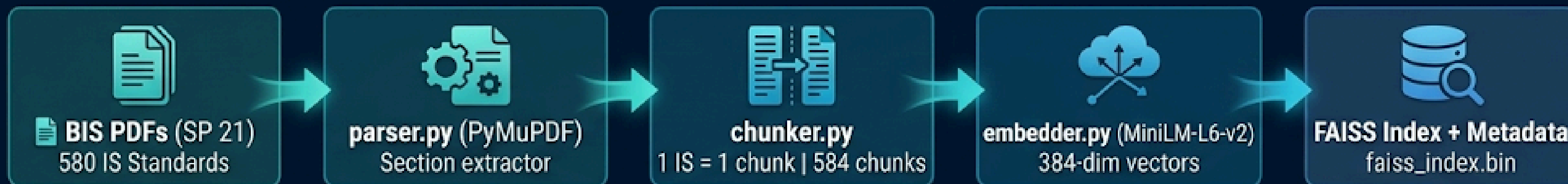
Natural Language Compliance in Seconds, Powered by Optimized RAG



Independent research tool. Not affiliated with BIS.

## 03 | System Architecture

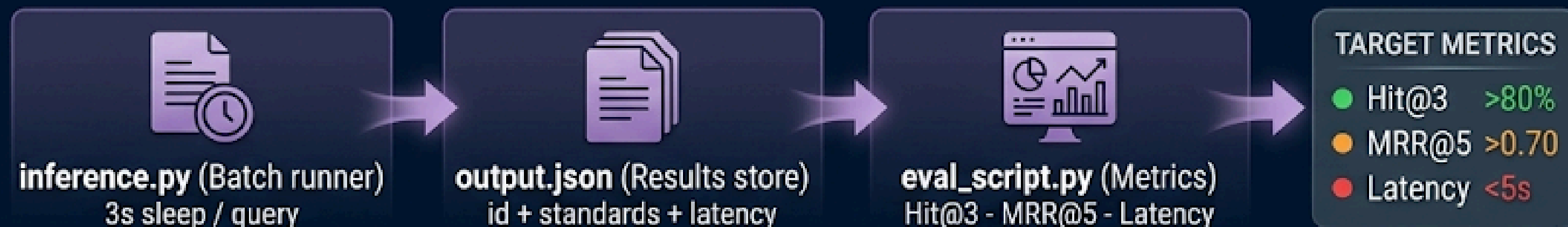
### PHASE 1 — OFFLINE INGESTION



### PHASE 2 — ONLINE INFERENCE (Per Query)



### EVALUATION



# Chunking & Retrieval Strategy

584 Chunks · 580 IS Codes

## 4-Stage Ingestion Pipeline



580 IS Standards

584 Total Chunks

3 Standards Split

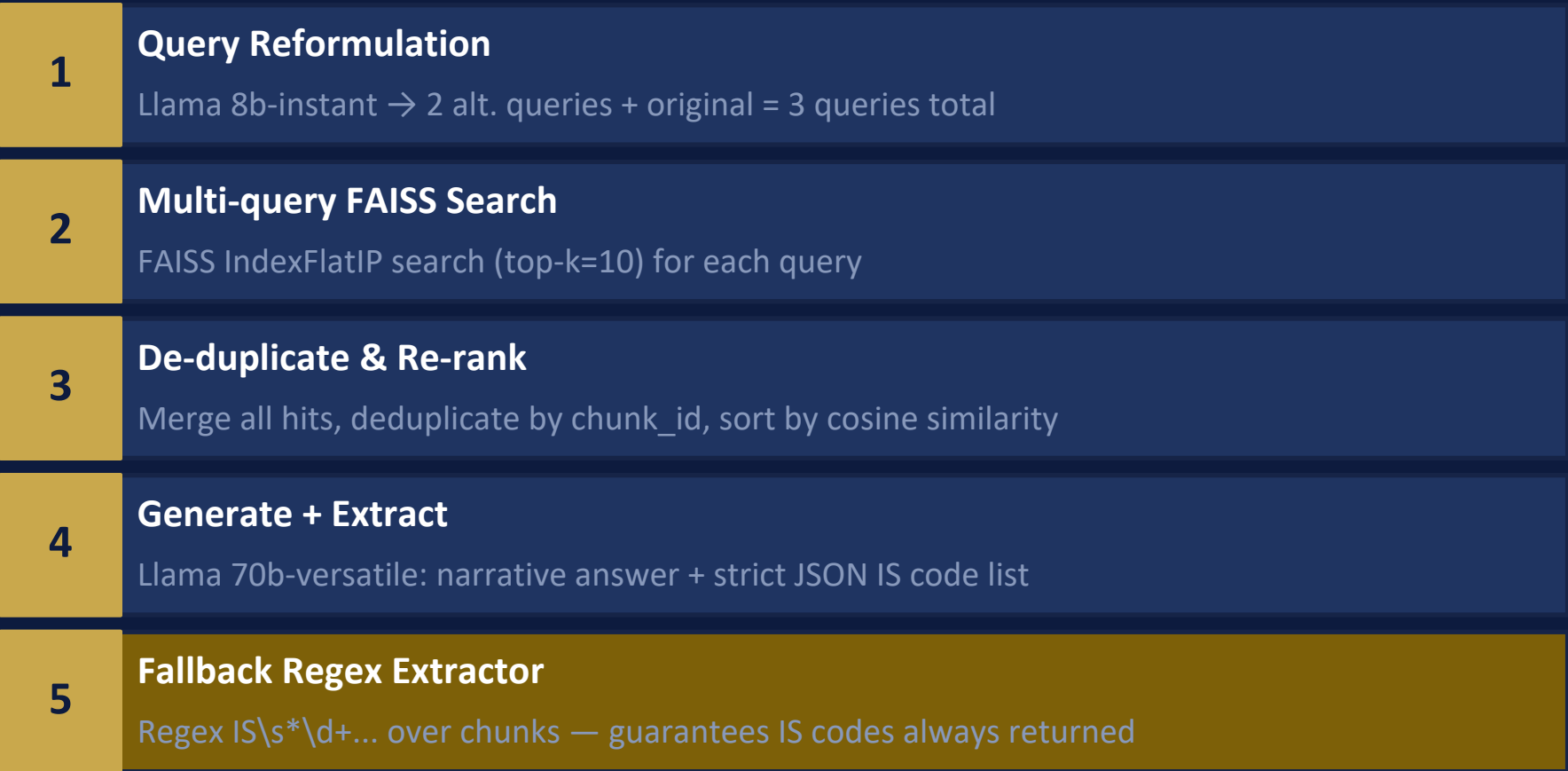
## Key Design Decisions

### Dual-Field Chunk Design

text\_to\_embed — Metadata header + content → fed to MiniLM encoder  
5-line header: IS Number · Title · Section · Source

content\_only — Scope + Requirements only → fed to LLM generation  
Saves ~180 tokens of LLM context per query (6 chunks × 30 tokens)

## Online Retrieval Flow



384-dim Embeddings

IndexFlatIP · Exact Search

~877 KB FAISS Index



# 05 | Demo Highlights

## Illustrative Example — Cement Manufacturer

User Input

Product: OPC 53 Grade Cement

Use: Structural concrete, RCC

Role: Manufacturer, BIS license pending

Reformulated Queries

Q1: "OPC 53 grade cement compressive strength IS specifications"

Q2: "Portland cement BIS license CRS mandatory requirements"

Q3: "cement fineness setting time soundness test IS 269"

### Top Recommendations Returned:

| IS 269:2015                                                                                                                                                        | IS 1489 Pt.1                                                                                                                                                          | IS 4031                                                                                                                                                             |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <div><div>Ordinary Portland Cement</div><div><div>Mandatory: Yes (CRS)</div><div>Confidence: High</div></div><div>→ Apply via BIS Simplified Procedure</div></div> | <div><div>Portland Pozzolana Cement</div><div><div>Mandatory: Yes (CRS)</div><div>Confidence: Medium</div></div><div>→ Apply via BIS Simplified Procedure</div></div> | <div><div>Methods of Physical Tests</div><div><div>Mandatory: Voluntary</div><div>Confidence: High</div></div><div>→ Apply via BIS Simplified Procedure</div></div> |

\* Placeholder output for illustrative purposes

# Evaluation Results

BIS Hackathon — Public Dataset

10

Queries Evaluated



Hit Rate @3

90%

Target >80%

✓ EXCEEDED

9 of 10 queries returned  
a correct IS code in top-3



MRR @5

0.90

Target >0.70

✓ EXCEEDED

Mean Reciprocal Rank  
across top-5 retrieved results



Avg Latency

1.39s

Target <5 sec

✓ EXCEEDED

End-to-end wall-clock time  
per query (incl. LLM calls)



# 07 | Impact on MSEs

From weeks of confusion to minutes of clarity.

## BEFORE BISfit

- ❌ 6–8 weeks searching BIS website manually
- ❌ Consultant fees of ₹50,000–₹2,00,000
- ❌ Risk of applying wrong / outdated standard
- ❌ No clarity on mandatory vs. voluntary
- ❌ No actionable next step for certification



## AFTER BISfit

- ✅ Standard recommendations within few seconds
- ✅ Zero consultant dependency for initial scoping
- ✅ Grounded strictly in official BIS documents
- ✅ Mandatory / CRS flag shown per recommendation
- ✅ Direct certification pathway mapped per standard

*BISfit bridges the compliance knowledge gap for 63 million MSEs — making BIS standards accessible, actionable, and fast.*